

1. Did you like your program topic? Why (not)?
 - a. I really enjoyed our program topic. It really reminded me of some of the games I used to code in Scratch as a kid. I think the features that Pygame provides are really fun to work with. I am glad the rules pushed us to work with something we have not used before, and I think it opened us up to more ideas for our game.
2. Did your team stick with your initial bottom-up approach to program development, or did you switch to a different design approach at some point? Which approach did you prefer and why?
 - a. Our team decided not to stick with the initial bottom-up approach. We did not have a lot of re-use of code within our functions, and felt like it was easier to plan from top to bottom. From starting broad with importing libraries and functions to defining each little object and circumstance in the code, it felt more natural to use a top-down approach to plan it all out.
3. Did your final design closely match the original hierarchy your team developed? In what ways is it the same? In what ways is it different?
 - a. Our final design did closely match the original design. We opted not to go with the bottom-up approach as our functions did not really overlap too much. This was the correct approach, as our initial design was pretty helpful for the creation of our program, and it ended up being pretty close to what we ended up with. I think if we went for the bottom-up approach, the design would definitely not have been very close at all.
4. Briefly describe how you used AI tools to complete your program. In what ways did AI tools help or hinder your progress?

- a. AI was used in our program to create the spritesheets and duck animations. It was also helpful for creating hitboxes for the different sprites, which were needed to determine if collisions occurred and also to track the collisions. AI was overall helpful as it was not the main source of developing our game, but was rather used to supplement anything we really struggled with, and made for a smoother process for developing this game.
5. Briefly describe the thing(s) you learned on your own, beyond what was covered in the lectures. How did you incorporate them into your program?
 - a. The things I learned on my own were implementing sprites and objects into the program and having them interact. It turns out that getting images and sprites from their respective files and implementing them into the code was very similar to CSV or TXT files. Additionally, I learned how to create hitboxes using rect as well as actually animate the sprites/objects used in our game.
6. If you had more time to work on your program, what additional features would you want to add? Is there anything about your submitted program you would change?
 - a. If we had more time, I would want to add a timer and a high-score tracker for each player who attempts the game. I think it would be cool to write to a file that tracks the top three users' stats and display them before each new run to help set a goal for the next user. It also creates a more competitive game and makes it more worthwhile to "grind."
7. What was the most difficult part of this assignment? Please explain.

- a. I think the most difficult part of the assignment was learning the different quirks of Pygame. There are a lot of things that are required with Pygame to get the desired outcomes we need, but we did not know how to use it very well. It was also really difficult to divide up the work and catch everyone up to speed with learning about Pygame in order to accomplish what they were tasked with.
- 8. Estimate the portion of the assignment completed by each member of your team (yourself included). Please explain any significant workload imbalances and give a brief summary of who did what.
 - a. Kai: 45% - Completed most of the code and was in charge of most of the development of the project. Had a lot of input on what functions to use, and overall had the most experience with Pygame.
 - b. Aidan: 35% - Completed the entire planning portion of the document. This included creating the initial hierarchy for the program, coming up with possible test cases, and explaining how the game should flow and overall work.
 - c. Jimmy: 10% - Helped to generate ideas for the game, but was busy studying for finals to truly make a large impact.
 - d. Simon: 10% Helped to generate ideas for the game, but was busy studying for finals to truly make a large impact.